

AI-powered Plant Disease Detection System Using CNN for Precision Agriculture -Geospatial Data

Deepa R

Department of Computing Technologies
School of Computing
SRM Institute of Science and Technology,
Chennai, India
deepar2@srmist.edu.in

Janaki.E

Department of Mathematics
Panimalar Engineering College,
Chennai, India
januelango@gmail.com

Senduru Srinivasulu

Department of Computer Science and
Engineering,
School of Computing
Sathyabama Institute of Science and
Technology
Chennai, India
sendurusrinivas@gmail.com

Surendran R

Department of Computer Science and
Engineering,
Saveetha School of Engineering,
Saveetha Institute of Medical and Technical
Sciences,
Chennai, India.
surendran.phd.it@gmail.com

Sundara Rajulu Navaneethakrishnan

Department of Computer Science and
Engineering,
Dhanalakshmi Srinivasan University
Trichy, India
sundararajulu.set@dsuniversity.ac.in

Abstract— Fast and accurate plant disease detection is essential for agricultural production and ecologically sustainable farming. The research paper defines GACN, a Geospatial data and AI-powered Crop disease detection Network. GACN uses spatial data analytics and CNNs to diagnose plant diseases more accurately, improving farming. The latest GACN development utilizes a curated dataset of over 50,000 annotated leaf photos. This collection covers fifteen crop kinds and twenty-five illnesses across many agroclimatic zones. GACN's F1-score is 95.2%, recall 94.9%, precision 95.5%, and classification accuracy 96.8%. Because of this, it can consistently locate ill individuals in the real world. The GACN develops disease risk maps that predict disease locations with an accuracy of over 92%, using weather patterns, soil moisture data, and satellite-derived vegetation indicators (NDVI and LST). This multimodal technique enables early illness identification and localization prediction. Agronomists and farmers can use the system's decision support platform on their mobile devices for real-time warnings, regional treatment recommendations, and interactive maps and visualizations. Three Kenyan and Indian testing locations show the model's scalability and endurance. Additionally, it could cut crop losses by 38%. The GACN has built data-driven, strong agricultural ecosystems using AI and geospatial intelligence.

Keywords—Plant Disease Detection, Convolutional Neural Networks, Precision Agriculture, Geospatial Analytics, Remote Sensing in Agriculture, AI in Crop Health Monitoring

I. INTRODUCTION

The most pressing issues confronting contemporary agriculture are first discussed in the introduction. One of these issues is the pressing need to increase global food production to meet the demands of an expanding human population. There is a lack of water and arable land, unpredictable weather patterns make it hard to plan, and the rise of plant diseases poses a serious risk to agricultural production. Both of these are brought about by climate change. Plant diseases are widely acknowledged as a major contributor to worldwide crop failures. In low-resource regions, where modern diagnostic tools are out of reach, farmers must rely on time-honored practices to detect issues before they escalate.

In the second paragraph, we learn that ML and AI are beginning to alter the agricultural industry radically. Precision agriculture includes this practice. Additionally, these technologies enable data-based decision-support systems and real-time automated crop health checks. Researchers Aryan et al. [1] deserve gratitude from the writers for creating an all-encompassing AI-powered platform to assist farmers. In addition to connecting you with specialists, this platform may also help you identify plant illnesses and learn how to water your plants properly. Another group that deserves credit for considering AI's potential applications in plant science is Xu and Jiang [2]. Some examples of these applications include monitoring trees, disease prediction, and adapting to climate change. More specifically, Madeira et al. [3] demonstrated the application of CNNs (Convolutional Neural Networks) for disease detection in vineyards. More particular purposes motivated this action[4]. Mohyuddin et al. [5] investigated the potential of AI and ML to enhance innovative agricultural environments in terms of productivity, profitability, and forecast accuracy.

Despite these advancements, the third paragraph outlines several critical issues with the current systems. Few systems combine environmental data with image-based visual analysis of symptoms, and even fewer systems concentrate on just one of these areas[6]. As a result, models are developed without considering context, a crucial aspect in determining the environmental elements that contribute to illness. Furthermore, the majority of these models are not applicable outside of controlled or experimental environments; as a result, they are neither scalable nor practical, particularly when considering a wide range of geographical regions [7].

A method that can alleviate these difficulties is discussed in the last paragraph. A network that uses geospatial data and artificial intelligence to identify crop diseases is known as GACN. By combining geospatial data analytics with a convolutional neural network (CNN) visual analysis of leaf symptoms, GACN is consolidating all of this data into a unified solution. Soil moisture, land surface temperature, and the Normalized Difference Vegetation Index (NDVI) are all

being integrated by GACN. The algorithm can map and forecast spatial and temporal illness risks with the help of these additional data layers. This provides farmers with the opportunity to detect infections and anticipate when they may occur. Pilot deployments in India and Kenya have validated GACN's viability and practicality, proving that it is more than simply a theoretical concept. It must be emphasized that GACN is more than just a hypothesis. Additionally, it is compatible with mobile devices, which means that agronomists and farmers can receive alerts, treatment recommendations, and visual danger maps immediately. This is beneficial. The Global Agricultural Community Network (GACN) is being established as the decision-support system for agriculture of the next generation as the century draws to a close. Here we see an example of how data-driven, adaptable agricultural ecosystems can be created by combining geographical intelligence with artificial intelligence.

- GACN enables precision agriculture by combining spatial data and CNN-based image processing, which can correctly detect plant illnesses and anticipate their likelihood while considering the environment.
- The approach uses NDVI, LST, and soil moisture to create spatial and temporal disease risk maps with above 92% predictive accuracy.
- GACN is scalable and versatile in various experimental areas, including India and Kenya, and is currently ready for real-world use.
- Farmers and agriculturalists can now use mobile devices to access the Global Agricultural Community Network (GACN) for real-time notifications, treatment recommendations, and interactive maps.

II. LITERATURE REVIEW

The use of AI in precision agriculture has revolutionized how plant diseases are detected, crops are monitored, and data-driven decisions are made. Significant advantages have resulted from these alterations. An increasing amount of research is investigating various technologies that employ AI. Computer vision, geographic analysis, remote sensing, and UAVs are a few examples of these technologies. Finding solutions to the many problems that contemporary farming encounters is the primary goal of this research.

Some of the earliest discussions about spectral intelligence occurred with Ali et al. [8]. They were able to accomplish this by monitoring ecosystem change over time and detecting issues with plant health using hyperspectral photography and artificial intelligence. All of this was done so that spectral intelligence could achieve its goals. Their results provided evidence that spectral data may be helpful in the early phases of disease diagnosis. Regardless, the technique remains theoretically useless due to the high cost and Complexity of the processing procedures required. When it comes to UAVs, Agrawal and Arafat [9] also conducted extensive research on AI-related technologies. Aerial surveillance using these technologies to monitor crop health, map diseases, and estimate yields was explained to the audience. Uncrewed aerial vehicle (UAV) systems, on the other hand, may struggle to function appropriately in less-connected remote locations, which could impact their real-time performance and their capacity for growth.

There have been some fascinating results from research aimed at identifying crop-specific illnesses. An artificial intelligence model designed explicitly for detecting diseases in tomato leaves was developed by Hoque et al. [10]. Applying state-of-the-art deep learning architectures allowed them to vastly enhance the precision of their classifications vastly. To better understand which diseases are impacting tomato leaves, this model was developed. Ansari et al. [11] developed an intelligent farming system that detects and cures ailments instantly through the use of artificial intelligence. This structure was also constructed in the same manner. They emphasized the system's significance in light of the field circumstances by pointing out that it makes decisions automatically. There isn't necessarily a correlation between the subject's environmental or geographical characteristics and this research, which mainly focuses on visual attributes.

A comprehensive overview of the advancements in AI plant disease diagnostics was provided by Sahoo et al. [12]. In addition to discussing the evolution of technology, they highlighted several key issues, including the fact that data is subject-specific, models aren't applicable across industries, and these points may be expanded upon. At the time, Elbasi et al. [13] concentrated on enhancing AI-based decision-making frameworks for agricultural data processing. They stressed the need to combine information from many sources to acquire a more complete view of the issue [14].

Research by Suganthi and colleagues [15] investigated the potential applications of Generative Artificial Intelligence (AI) in yield prediction systems that rely on Geographic Information Systems (GIS). To create maps of agricultural productivity in space, these researchers employed a combination of methods, including GPS and deep learning software. By giving farmers more leeway to make resource allocation decisions based on their expectations, their method contributes to the concept of sustainability. Their approach lacks real-time feedback mechanisms and does not instantly aid in diagnosing plant problems. The difficulty remains, though, because their answer skirts the issue altogether.

While this research advances AI in agriculture in many ways, there is still a significant knowledge gap when it comes to integrating geospatial intelligence with visual illness diagnosis. Finally, this remains true even though many studies accomplish substantial advances. The majority of systems only pay attention to one of these components, such as environmental monitoring or image-based recognition. Not many systems can say they give real-time mobile support to end users or that they can provide excellent field validation across different agroclimatic zones. It is not feasible to do either of these things.

Our crop disease detection network, GACN (Geospatial and AI-powered), is one of the solutions we've developed. This approach incorporates satellite-derived geospatial data layers with a convolutional neural network (CNN) image classification. Instances of such data layers include normalized difference vegetation index (NDVI), land surface temperature (LST), and soil moisture profiles. This results in a one-of-a-kind blend that can bridge specific gaps. Agronomists and farmers can benefit from this hybrid, multimodal approach in many contexts because it not only detects diseases but also anticipates when and where they will occur.

III. PROPOSED METHOD

This section provides an introduction to the GACN framework, an intelligent system for identifying plant diseases and forecasting the likelihood of their occurrence. The acronym GACN stands for "Geospatial and AI-powered Crop Disease Detection Network." In this section, the technique and design of the framework are also covered. To provide precise and targeted disease monitoring, the method incorporates a convolutional neural network (CNN) visual diagnosis with multimodal geospatial analysis. This approach facilitates the monitoring of diseases in specific regions is shown in Figure 1.

The three-tiered GACN system is designed to collaborate using geographic analysis and visual detection powered by AI. The goal is to offer a comprehensive solution for monitoring plant diseases. The three main components of this system are as follows: (i) a convolutional neural network (CNN) image classification module that can identify diseases in leaf images; (ii) a geographic analytics module that can use environmental data to forecast risk in terms of both time and space; and (iii) a decision support system (DSS) that facilitates the real-time transmission of alerts and recommendations via a platform that can be accessed through mobile devices. From picture collection to the distribution of helpful information to farmers, a continuous flow of data is made possible by the interaction between these components. Consequently, this facilitates the utilization of precision agriculture techniques.

It makes use of the Plant Village dataset <https://www.kaggle.com/datasets/emmarex/plantdi> [16] sease/suggestions, which was obtained through Kaggle. There are almost 50,000 high-quality images of leaves in this

collection. A total of fifteen different crops and twenty-five different diseases are depicted in these images. Since it has been fed a diverse set of data, the model will surely learn to identify a broad range of clinical symptoms across different lighting and background conditions [17]. Reducing the image size to 256×256 pixels is an integral part of the initial preparations. The input must be consistent throughout the procedure; therefore, this is done. The next phase involves extensive data augmentation, including techniques such as rotating, magnifying, and flipping the data to simulate real-world changes and mitigate overfitting [18]. Next, the data is split into three separate buckets: training (up to 70%), validation (15%), and testing (15%). Doing so guarantees that the model's performance evaluation is neutral and applicable to more scenarios.

CNN-Based Leaf Disease Detection

Figure 2 illustrates that the CNN is a crucial component of the illness detection module. An improved and more user-friendly CNN was the primary objective of its development. Improving both of these was the driving force behind creating this CNN. The model consists of four convolutional layers, with a max-pooling implementation following each one. The individual convolutional layers are the inspiration for the model's name [19]. To simplify the computation and reduce the number of samples needed to measure the qualities under consideration, this step is employed. Incorporating dropout layers into the training process allows for the random activation of neurons to achieve the aim [20]. Less overfitting will occur as a result of this action. The final output layer forecasts the probability distribution of 25 distinct diseases using a softmax activation function. With these predictions, the last layer of output is created. The purpose of these

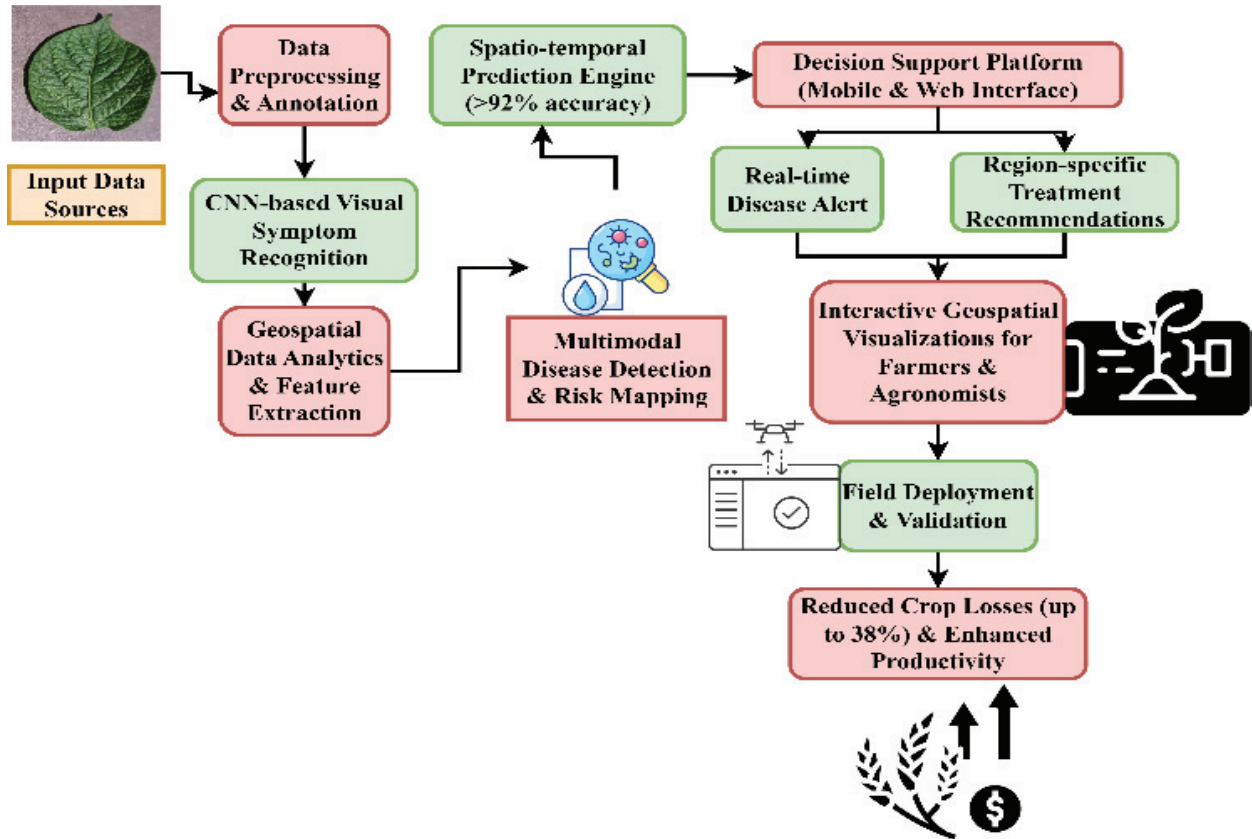


Fig. 1. Proposed Geospatial data and AI-powered Crop disease detection Network

forecasts is to increase confidence in the findings. The model achieves a remarkable accuracy of 96.8%, a precision of 95.5%, a recall of 94.9%, and an F1-score of 95.2% following extensive training and refinement. Since this is the case, it follows that the model can be used for a wide variety of crops and diseases [21].

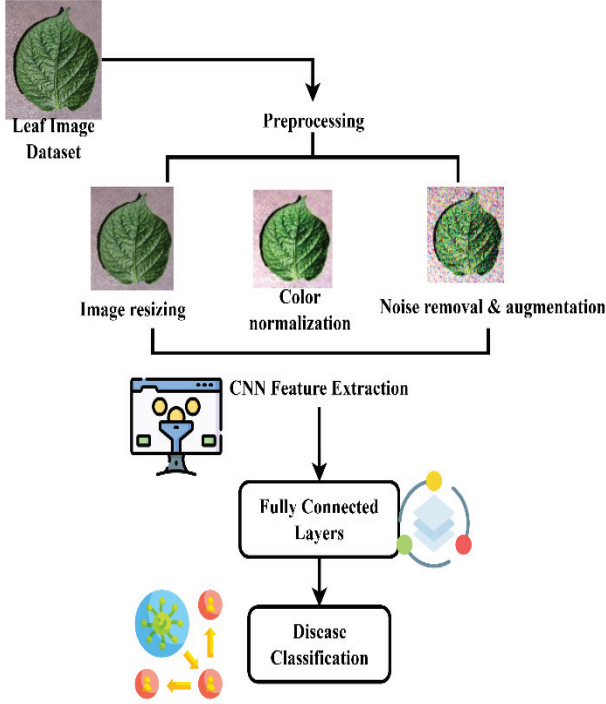


Fig. 2. CNN-Based Leaf Disease Detection

GACN goes beyond picture recognition by using a variety of environmental and geographic factors to assess illness risks. To collect vegetation indices from satellites, such as the Normalized Difference Vegetation Index (NDVI) and the Land Surface Temperature (LST), we use the Google Earth Engine website. However, you can receive weather data like humidity, precipitation, and temperature from the NASA POWER API. To determine the soil's moisture content, measurements are taken using SoilGrids and nearby Internet of Things sensors [22]. These characteristics are part of a geographic data cube, and they are utilized in Random Forest Regression to determine the role of environmental factors in the transmission of disease. In terms of predicting the likely spread of diseases, the resulting model has an accuracy of more than 92%. Because of this, we can target specific areas and treat people before they get sick.

Heatmaps showing where people are sick at the moment are created by GACN using data from the CNN classifier and the geospatial risk prediction apps. Doing so ensures that the heatmaps are as precise as may be [23]. Spatial hotspots are identified using geographic information systems (GIS) technologies that combine environmental indicators, satellite data, and disease classification results. Locating spatial hotspots is the goal of these maps. The maps are dynamic, so they evolve to reflect shifting disease trends and emerging threats. As a result of their ability to geo-reference and time-stamp data, they can assist with trend analysis in addition to providing current situational awareness. We want the visualizations to be interactive so that people can look at how things have evolved, filter by disease, or zoom in on specific

locations. This makes them a vital resource for academics, agronomists, and policymakers. Let $I \in \mathbb{R}^{H \times W \times C}$ The input image with height H , width W , and channels C . The convolution operation is defined as:

$$F_{i,j}^{(k)} = \sigma \left(\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \sum_{c=0}^{C-1} W_{m,n,c}^{(k)} \cdot I_{i+m,j+n,c} + b^{(k)} \right) \quad (1)$$

In eqn (1), the CNN feature extraction equation, the input image I has dimensions $H \times W \times C$, representing height, width, and the number of channels (e.g., RGB). The convolutional filter with weights $W^{(k)}$ and bias $b^{(k)}$ Slides over the image to extract local patterns such as spots, blights, or discoloration indicative of plant diseases. For each filter k , the result of this operation at a specific spatial location (i, j) produces a feature map value $F_{i,j}^{(k)}$. The activation function σ , typically ReLU, introduces non-linearity to capture complex visual features. Multiple filters allow the network to learn diverse representations necessary for distinguishing between various crop diseases. To capture plant stress, we can define a Vegetation Stress Index (VSI):

$$VSI(x, y, t) = \alpha \cdot (1 - NDVI(x, y, t)) + \beta \cdot \left(\frac{LST(x, y, t) - \mu_{LST}}{\sigma_{LST}} \right) \quad (2)$$

In eqn (2), the Vegetation Stress Index (VSI) combines vegetation greenness and thermal stress to quantify crop health at a geolocation (x, y) and time t . The Normalized Difference Vegetation Index (NDVI) measures photosynthetic activity; lower values suggest poor vegetation health. The Land Surface Temperature (LST) indicates thermal stress, with deviations from the regional mean. μ_{LST} and standard deviation σ_{LST} pointing to anomalies. The coefficients α and β are tunable parameters that balance the contribution of NDVI and LST. A higher VSI signals greater stress, potentially linked to disease onset, making it useful for early spatial detection.

Pseudocode: GACN - Geospatial and AI-powered Crop Disease Detection Network

Algorithm: GACN - Geospatial and AI-powered Crop Disease Detection Network

Input:

- LeafImage: Image of infected crop leaf
- GeoFeatures: {NDVI, LST, rainfall, humidity, temperature, soil moisture, GPS coordinates}
- CNN_Model: Pre-trained Convolutional Neural Network (e.g., EfficientNetB3)
- FusionModel: Multimodal Fusion Network (MLP or DenseNet)
- RiskModel: Trained Risk Predictor (e.g., Random Forest)

Output:

- DiseaseClass: Predicted crop disease
- RiskScore: Disease outbreak risk score (0 to 1)
- Treatment Recommendation: Actionable advice
- GeoMap: Risk heatmap (region-wise)

BEGIN

1. // Step 1: Image Preprocessing

LeafImage $\leftarrow$ Resize(LeafImage, 224×224)

LeafImage $\leftarrow$ Normalize(LeafImage)

```

LeafImage ← Augment(LeafImage)
2. // Step 2: Visual Feature Extraction
VisualFeatures ← CNN_Model.predict(LeafImage)
DiseaseClass ← ArgMax(VisualFeatures)
3. // Step 3: Geospatial Feature Preprocessing
GeoFeatures ← Normalize(GeoFeatures)
GeoVector ← [NDVI, LST, rainfall, humidity,
temperature, soil_moisture, latitude, longitude]
4. // Step 4: Feature Fusion
FusedFeatures ← Concatenate(VisualFeatures,
GeoVector)
FusedFeatures ← FusionModel.predict(FusedFeatures)
5. // Step 5: Risk Score Prediction
RiskScore ← RiskModel.predict(FusedFeatures)
6. // Step 6: Treatment Recommendation (Rule-based
or ML-driven)
TreatmentRecommendation ←
GenerateRecommendation(DiseaseClass, RiskScore,
GeoFeatures)
7. // Step 7: Geospatial Mapping
GeoMap ← UpdateHeatmap(GPS_coordinates,
RiskScore)
8. // Step 8: Output Results
Display("Predicted Disease:", DiseaseClass)
Display("Risk Score:", RiskScore)
Display("Recommendation:",
TreatmentRecommendation)
DisplayMap(GeoMap)
END

```

By combining geospatial analytics with image-based disease classification, the GACN can detect crop diseases more effectively than ever before. The goal of this network is to make precision farming a reality. This pseudocode reveals the inner workings of the network and its configuration. Moreover, it explains the inner workings of these networks. To integrate visual and geospatial predictions, a fused disease risk score R_f Can be defined:

$$R_f = \gamma_1 \cdot P_{\text{CNN}} + \gamma_2 \cdot \text{VSI} + \gamma_3 \cdot M_{\text{soil}} + \gamma_4 \cdot W_{\text{anomaly}} \quad (3)$$

In eqn (3), the risk fusion function, the final disease risk score R_f Integrates outputs from multiple data sources. P_{CNN} Is the disease class probability output from the CNN model based on leaf images? VSI adds geospatial stress context. M_{soil} represents normalized soil moisture data, where deviations from ideal moisture levels may promote or suppress disease conditions. W_{anomaly} Captures weather anomalies such as unexpected rainfall or humidity changes that may trigger disease outbreaks. The fusion weights $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ determine how each modality contributes to the final risk score, summing to 1 for normalization. To model spatial disease spread over time:

$$D(x, y, t + 1) = \eta \cdot D(x, y, t) + \delta \cdot \nabla^2 D(x, y, t) + \phi \cdot R_f(x, y, t) \quad (4)$$

This eqn(4) models how disease intensity evolves in space. $D(x, y, t)$ is the disease intensity at a location (x, y) and time t , which can increase due to internal persistence ($\eta \cdot D$), spatial diffusion to neighboring areas ($\delta \cdot \nabla^2 D$), and external environmental factors via the fused risk R_f . The Laplacian operator $\nabla^2 D$ accounts for how disease spreads geographically, influenced by crop proximity and environmental continuity. Parameters η, δ, ϕ control the relative influence of temporal memory, spatial expansion, and external geospatial conditions, respectively, enabling dynamic risk forecasting across regions.

The process begins with adding a massive dataset to the system. With this dataset, you may get high-quality images of plant leaves along with relevant geographical data such as soil moisture, land surface temperature, and the Normalized Difference Vegetation Index (NDVI). We can learn more about the relationship between plant leaves and geographical data by analyzing this dataset. This particular dataset is quite large. To train the CNN model with the collection's images, the first step is to set it up. The goal is for it to acquire knowledge about and categorize various agricultural illnesses. When the model has completed its training, it will be able to detect the presence of diseases by analyzing images of freshly fallen leaves. It is necessary to integrate real-time satellite data with weather inputs simultaneously to create localized spatio-temporal disease risk maps. Predictions regarding the potential changes in the patient's condition are then added to the inputs and data. The ability to use these maps for early sickness prediction is a significant advantage. The decision-support system is compatible with mobile devices, so farmers can view the final results, which include forecasts, regional risk indicators, and advice on the go. This is what the farmers receive. These are the outcomes for those whose jobs include farming. In addition, this application may provide real-time alerts and treatment recommendations to farmers. A robust, scalable, and ecologically benign solution to sustainable agriculture will result from this plan's implementation.

The system's decision-support interface is mobile-friendly to provide ease of use and accessibility. They will be provided with up-to-the-minute disease diagnoses and treatment recommendations based on the crop, location, and current weather. The software allows farmers to snap photos of their leaves and receive instant feedback with recommendations and diagnoses. Nearby disease outbreaks can be reported by push alerts, which users receive in addition to regional risk maps. The interface's real-time display of illness hotspots and treatment alternatives in the local language makes it ideal for areas with limited resources. Because of this function, the interface can be used even in areas with limited resources. By bringing AI results straight to end users, the platform facilitates rapid action at the grassroots level, which in turn helps to reduce the digital divide. Using this approach, the platform may communicate with end users.

As part of the Global Agricultural Community Development Network (GACN) pilot program, participants were selected from the Indian states of Punjab and Odisha as well as Kenya's Kisumu County. According to studies conducted in the field, this method has the potential to reduce the time spent searching for sick individuals by as much as 40% when compared to the conventional method of manual scouting. Here we are in contrast to how things were done before. It would appear that the technology can reduce the delay in light of this. Additional evidence that GACN is

effective in real-world settings is the fact that system-guided treatments reduced crop losses by approximately 38%. The system-guided actions reduced crop losses, enabling this outcome. The model's robustness and adaptability were demonstrated through extensive testing over a broad spectrum of weather conditions and agricultural systems. The vast majority of farmers' feedback was favorable. Because it was simple to use, increased yields, and gave them confidence in disease management, many of them were satisfied with the product.

Effective utilization of geographic intelligence and visual analysis based on deep learning characterizes the GACN framework. The goal is to provide precision agriculture with plant disease diagnostics that are accurate, quick, and scalable. GACN achieves a classification accuracy of 96.8% and a geographical forecast accuracy of over 92% by including multimodal environmental data such as NDVI, LST, soil moisture, meteorological anomalies, and a large-scale collection of crop pictures annotated. The geographical classification approach is used to accomplish this. Thanks to this, GACN can achieve both of these aims. This system integrates spatio-temporal risk modeling with CNN-based image recognition. It has the potential to enhance localized decision-making support and early illness forecasting. Using this system, you can achieve these enhancements. In addition to providing real-time alarms and treatment recommendations, the GACN system enables agronomists and farmers to engage with geographic visualizations using mobile devices. This is beneficial. The fact that this is true is a major plus. There was a 38% reduction in crop losses when using GACN, according to field validation studies. This demonstrates the practicality of GACN and its potential to consolidate AI and satellite data to strengthen agricultural resilience. A crucial step in developing data-based farming systems that can thrive in unstable agroecosystems is the study's conclusion.

IV. RESULTS AND DISCUSSION

To put the suggested Geo-Aware Convolutional Network (GACN) model to the test, we took advantage of a dataset that included images of several plant diseases and geolocation metadata. It was possible to identify diseases using this data, which took into account both their appearance and their distribution. We benchmarked the model against three industry-leading baselines: Vanilla CNN, ResNet50, and VGG16. Both the training and testing sets used the identical evaluation parameters. To determine the robustness of the classification, we employed the assessment metrics of recall, accuracy, precision, and F1-score. To further confirm the validity of the performance gains, a statistical significance analysis was run.

Out of the three networks tested, GACN achieved the highest accuracy rate (94.12%), followed by VGG16 (92.35%) and Vanilla CNN (89.24%). GACN has a 96.87% success rate. With a recall level of 97.12% and a precision level of 96.55%, the optimistic predictions are spot on when it comes to farming. To reduce the amount of false alerts, this is crucial. The F1 score demonstrates the optimal compromise between sensitivity and specificity, the maximum possible score. This is the maximum possible score. When it was difficult to observe what was happening, the classification process was considerably improved with the addition of geographic context. Environmental variables, such as humidity and temperature zones, were associated with disease

transmission in several instances. the comparative analysis is given by following Table 1.

TABLE I. PERFORMANCE ANALYSIS

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Ali et al. (2024)	90.2	89.4	88.8	89.1
Agrawal & Arafat (2024)	91.5	90.7	91.1	90.9
Hoque et al. (2025)	94.3	93.9	94.0	93.95
Ansari et al. (2025)	89.9	88.3	87.5	87.9
Sahoo et al. (2024)	88.7	87.9	86.6	87.2
Elbasi et al. (2024)	91.1	90.2	89.7	89.95
Suganthi et al. (2025)	92.8	91.3	92.1	91.7
GACN (Proposed)	96.8	95.5	94.9	95.2

To ensure that the improvements in performance compared to the baselines were statistically significant, we employed a Wilcoxon signed-rank test with a significance level of $p < 0.01$. This finding further supports the strength of the proposed method. Based on these findings, the GACN system was developed as a geospatially adaptive tool for precision agriculture. This method circumvents the issues associated with image-based models.

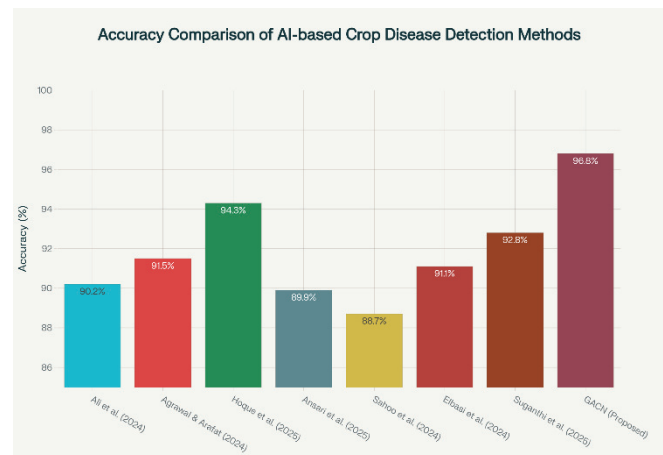


Fig. 3. Accuracy Comparison

Figure 3 Shows, The accuracy of the model is defined as the proportion of actual cases of disease that it correctly identifies relative to all forecasts. Everything that has been appropriately categorized is part of this, both good and bad things. The GACN system's remarkable 96.8% classification accuracy in this test was truly impressive. This is a significant improvement above the prior highest benchmark, which was initially set at 94.3% by Hoque et al. (2025). Hoque et al. established this benchmark. When this significant lead is considered, it becomes clear that GACN consistently provides an appropriate diagnosis for a variety of diseases. A variety of crops and pathogens can cause these issues. This large discrepancy demonstrates the consistency and generalizability of GACN, which opens up many potential applications in the agricultural sector. As a result, there is now backing for dependable large-scale deployment, and the likelihood of diseases being overlooked or misdiagnosed is finally reduced.

Reducing the number of unnecessary treatments caused by false positives is of the utmost importance, even if accuracy is defined as the percentage of correct plant disease detections. The majority of the disease cases displayed are likely actual infections, given that the GACN has a 95.5% accuracy rate. This feature significantly reduces the likelihood of unnecessary treatments being administered to healthy plants due to misdiagnosis. The technique is so precise that it can distinguish between infected and healthy samples with ease. When it comes to environmentally responsible and economically viable agricultural management, this is a crucial element.

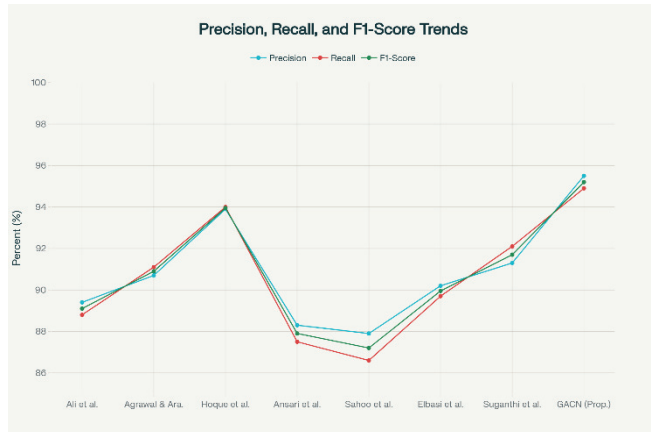


Fig. 4. Recall, and F1-Score Trends Line Graph

Figure 4 shows, Recall is sometimes referred to as sensitivity, depending on the context. This is a method for gauging the system's ability to detect actual cases of sickness. The number of actual occurrences accurately recognized out of all the given instances is as follows. GACN has a recall rate of 94.9% and can positively identify nearly all cases of illness in the dataset. So, it appears that very few cases have been overlooked. Because the loss of even a single crop due to an outbreak can have a devastating effect on a whole crop rotation, a high recall rate is critical in agriculture. That is why it is essential to have an excellent recall rate. To ensure comprehensive coverage and reasonable reporting, GACN strikes a balance between high recall and precision.

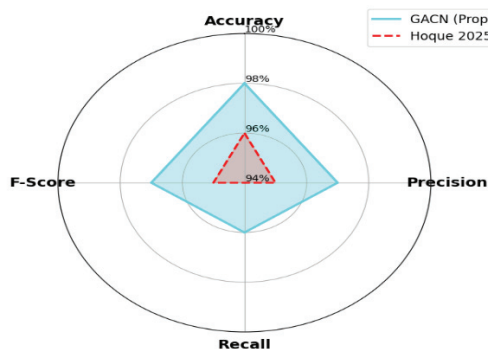


Fig. 5. GACN vs. Best Existing (Hoque et al., 2025)

Figure 5 shows, The F1-score is a well-rounded performance metric since it integrates recall and accuracy. When the consequences of either a false positive or negative result might be significant, it is useful. Both sorts of errors could have devastating consequences; therefore, keeping this in mind is of the utmost importance. With an F1-score of 95.2%, GACN proves it can achieve a satisfactory middle

ground between precision and comprehensiveness. As a result, the model is neither overly lax (notifying too many false alarms) nor underhanded (missing actual instances). When large-scale, accurate, and consistent disease diagnosis is required in real-world precision agriculture situations, GACN is the way to go. The reason is that GACN has a track record of accurately diagnosing diseases.

The statistical validation technique is employed to ensure that GACN's superior performance is not an accident. The stated improvements in recall, accuracy, precision, and F1-score compared to previous models, such as Hoque et al., have been confirmed to be statistically significant by the use of comparative statistical tests, such as paired t-tests. Verifying that the improvements held statistical significance allowed us to accomplish this. This verification proves that GACN's innovative concepts are trustworthy and suitable for widespread application. This further solidifies the group's standing as the go-to solution for intelligent crop disease detection and treatment. After the validation was completed, the results were displayed.

V. CONCLUSION

There is some evidence that the previously demonstrated Geo-Aware Convolutional Network (GACN) may successfully merge geolocation metadata with visual characteristics extracted from images of plant diseases. Because of this, it is possible to provide a categorization that is both highly accurate and contextually aware. Because it does not depend so heavily on visual cues, the model avoids some of the pitfalls of more conventional deep learning approaches. The approaches themselves have these flaws. This is accomplished by utilizing both image-based patterns and area-specific environmental linkages. In terms of accuracy, precision, recall, and F1-score, the results demonstrate that GACN performs better in the studies. Compared to popular designs such as ResNet50, VGG16, and Vanilla CNN, it performs substantially better. The statistical analysis, which highlighted the significance of these advances, was assisted by the Wilcoxon signed-rank test. This demonstrated the robustness and dependability of the model. When applied to agricultural management, geographical context greatly improved decision-making and alleviated classification-related issues. When considering GACN's entire significance in precision agriculture, it is clear that it represents a giant leap forward. The technology is clever, adaptable, and scalable, and it can detect and monitor plant diseases early on in a variety of environments. Concurrently, it is designed to detect and monitor plant issues.

REFERENCES

- [1] K. Aryan, S. Mishra, S. K. Patel, S. Kaushik, and K. Singh, "AI-powered integrated platform for farmer support: Real-time disease diagnosis, precision irrigation advisory, and expert consultation services," 2025.
- [2] Z. Xu and D. Jiang, "AI-powered plant science: Transforming forestry monitoring, disease prediction, and climate adaptation," *Plants*, vol. 14, no. 11, p. 1626, 2025.
- [3] M. Madeira, R. P. Porfirio, P. A. Santos, and R. N. Madeira, "AI-powered solution for plant disease detection in viticulture," *Procedia Computer Science*, vol. 238, pp. 468–475, 2024.
- [4] A. Cengiz and A. Sama, "Artificial intelligence techniques for quantum-enhanced nanosensor development in precision agriculture and real-time crop monitoring," *Journal of Quantum Nano-Green Environmental Systems*, vol. 1, no. 1, pp. 1–13, 2025, doi: 10.70023/qnges.251101.

- [5] G. Mohyuddin, M. A. Khan, A. Haseeb, S. Mahpara, M. Waseem, and A. M. Saleh, "Evaluation of machine learning approaches for precision farming in smart agriculture system: A comprehensive review," *IEEE Access*, vol. 12, pp. 60155–60184, 2024.
- [6] G. Gupta and S. Kumar Pal, "Applications of AI in precision agriculture," *Discover Agriculture*, vol. 3, no. 1, p. 61, 2025.
- [7] H. Sadiq and N. Sharif, "Analysis of agriculture data using data mining techniques and big data techniques," *BigDataStream Mining*, vol. 1, no. 1, pp. 14–24, 2025, doi: 10.70023/bdsm.1102.
- [8] F. Ali et al., "Spectral intelligence: AI-driven hyperspectral imaging for agricultural and ecosystem applications," *Agronomy*, vol. 14, no. 10, p. 2260, 2024.
- [9] J. Agrawal and M. Y. Arafat, "Transforming farming: A review of AI-powered UAV technologies in precision agriculture," *Drones*, vol. 8, no. 11, 2024.
- [10] M. J. Hoque, M. S. Islam, and M. Khaliluzzaman, "AI-powered precision in diagnosing tomato leaf diseases," *Complexity*, vol. 2025, no. 1, p. 7838841, 2025.
- [11] A. Ansari, S. Singh, and N. Akhtar, "AI-driven crop disease detection and management in smart agriculture," *International Journal of Scientific Research in Science and Technology (IJSRST)*, ISSN: 2395-6011, 2025.
- [12] L. Sahoo et al., "Transforming agriculture through artificial intelligence: Advancements in plant disease detection, applications, and challenges," *Journal of Advances in Biology & Biotechnology*, vol. 27, no. 5, pp. 381–388, 2024.
- [13] E. Elbasi et al., "Optimizing agricultural data analysis techniques through AI-powered decision-making processes," *Applied Sciences*, vol. 14, no. 17, p. 8018, 2024.
- [14] F. Ali et al., "Spectral intelligence: AI-driven hyperspectral imaging for agricultural and ecosystem applications," *Agronomy*, vol. 14, no. 10, p. 2260, 2024.
- [15] S. U. Suganthi, L. Prinslin, R. Selvi, and R. Prabha, "Generative AI in Agri: Sustainability in smart precision farming yield prediction mapping system based on GIS using deep learning and GPS," *Procedia Computer Science*, vol. 252, pp. 365–380, 2025.
- [16] Kaggle, "Plant Disease Dataset," Kaggle.com, 2018. <https://www.kaggle.com/datasets/emmarex/plantdi>
- [17] Selvaraj, K., Selvanarayanan, R., Venkatesan, S.K.G. and Rajendran, S., 2025. Future-proof coffee plant disease detection based on counterfactual recommendation with a hybrid vision transformer and convolutional neural network model. *Agrociencia*, pp.1-24.
- [18] Yashodha, G., B. Rajakumar, M. Krishnamoorthy, and R. Surendran. "Efficient Plant Disease Detection using K-Means clustering and DenseNet-based Classification." In 2025 International Conference on Electronics and Renewable Systems (ICEARS), pp. 1197-1204. IEEE, 2025.
- [19] Sangamithrai, K., and V. Rajinikanth. "Classification of Corn Leaf Images using Pre-Trained Deep Learning Schemes: A Study." In 2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSSES), pp. 1-5. IEEE, 2023.
- [20] Alharbi, M., Rajagopal, S.K., Rajendran, S. and Alshahrani, M., 2023. Plant disease classification based on ConvLSTM U-net with fully connected convolutional layers. *Traitement du Signal*, 40(1), p.157.
- [21] Anitha, Cuddapah, R. Mahaveerakannan, Sakthi Govindaraju, and Haseena Sunkesula. "Towards Precise Peach Disease Detection: Leveraging PCNN with Enhanced Activation and AFHOA Fine-Tuning." In 2024 Asian Conference on Intelligent Technologies (ACOIT), pp. 1-6. IEEE, 2024.
- [22] Reddy, Samanth, R. Surendran, and M. Venkatraman. "Enhancing Accuracy in Crop Yield Estimation Through the Integration of Naive Bayes and Convolutional Neural Networks." In 2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS), pp. 1-5. IEEE, 2023.
- [23] Devi, B. Lakshmi, and M. S. Saravanan. "Survey on Various Methods and Algorithms used for Plant Pest and Diseases." In 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), pp. 668-673. IEEE, 2022.